
Auditing a Deterministic Agent in Language-Based Economic Games*

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Abstract

We audit ELP-AGENT, a deterministic, role-separated entrant for GLEE bargaining, bilateral negotiation, and persuasion. The submitted policy uses no language model at inference time: numeric actions come from explicit opponent-path summaries and heuristic thresholds, while validators and a one-assignment controller constrain execution and exposure. Three development failures motivated the audit: a legal 99-round cycle, assignment overshoot, and family totals that concealed a losing role. Across the complete 1,704-game dashboard history, displayed changes sum to $+535.9/ +695.0/ +833.0$ for the three families, while sequential role comparisons show observed sign reversals even under unchanged code. In the final adaptively stopped trace, bargaining/persuasion/negotiation changed $+190.94/ +55.67/ -33.73$ over 150/78/20 games. An author-supplied next-day snapshot placed the entrant 95th overall at 1674.3. These outcomes neither estimate policy effects nor validate the persuasion seller’s static Bayes calibration: matchmaking was uncontrolled, policies were selected using earlier live runs, and two families were censored by drawdown guards. The contribution is therefore an executable policy specification and a failure-oriented case study of how strategic decisions, action validity, and exposure monitoring can diverge.

1 Scope and contributions

GLEE evaluates agents in bargaining, bilateral trade, and persuasion with optional natural-language interaction, heterogeneous horizons, and incomplete information [10]. We study a deliberately in-spectable entrant: its economic actions are produced by fixed Python rules rather than model inference. This makes failures attributable to explicit state summaries, branch logic, validation, or the run controller.

This paper makes three bounded contributions. First, it gives a self-contained specification of the six deployed role policies and their memory boundaries (Appendix A). Second, it audits failures that legal-action rate or aggregate rating alone would hide: cycling, uncontrolled exposure, role reversals, and a monitoring-order bug. Third, it releases the exact executable, ordered evidence, analysis code, and a 248-game sequential trace. We do *not* claim strategic superiority, causal mechanism effects, dynamic receiver obedience, or generalization beyond this run.

The methodological novelty is the alignment of three normally separate audit layers—strategic decisions, contract enforcement, and sequential exposure—on one immutable event trace. This decomposition reveals when legal execution, apparently safe monitoring, and rating success disagree; it is an audit design, not a new equilibrium or learning algorithm.

*Repository: <https://github.com/SyedT1/GLEE-Competition>.

2 Related work

GLEE joins economic games with language-agent evaluation [10]. Automated-negotiation research has long separated opponent preference, strategy, and deadline modeling [2]; learned dialogue agents instead optimize negotiation end to end [6]. Our explicit separation of numeric policy from fixed language resembles modular strategy-generation designs [3], but we contribute neither a learned dialogue model nor a new equilibrium solution. Interactive-agent benchmarks emphasize task-level success and failure [7]; internal algorithmic audits emphasize lifecycle evidence [8], while runtime shields enforce action-level specifications [1]. Our narrower contribution aligns strategic output, a contract validator, and an exposure controller against the same ordered trace; unlike a formal shield, its validator guarantees action legality rather than reward. The persuasion inequality below is a direct static Bayes constraint related to Bayesian persuasion [5], not a dynamic result. Confidence sequences address optional stopping [4], but this trace does not establish an estimand or the conditional dependence assumptions needed for a valid anytime analysis.

3 Submitted policy and safety layers

State and memory. The dispatcher locks a family–role branch for each game. The policy is deterministic conditional on the complete visible history, persistent memory, and game identifier—not on the current state alone. Bargaining memory stores opponent demands and rejected shares by role, information regime, discount configuration, and disclosed opponent; negotiation stores role-specific price paths; the persuasion buyer updates signal precision only when purchase reveals quality. Undisclosed opponents receive game-local keys. Messages verbalize an already selected action, although they may indirectly affect later states. Figure 1 shows the tradeoff optimized in bargaining, the price and response threshold in negotiation, and the constrained signal paths in persuasion. It locates Eqs. 1–3 within those decisions.

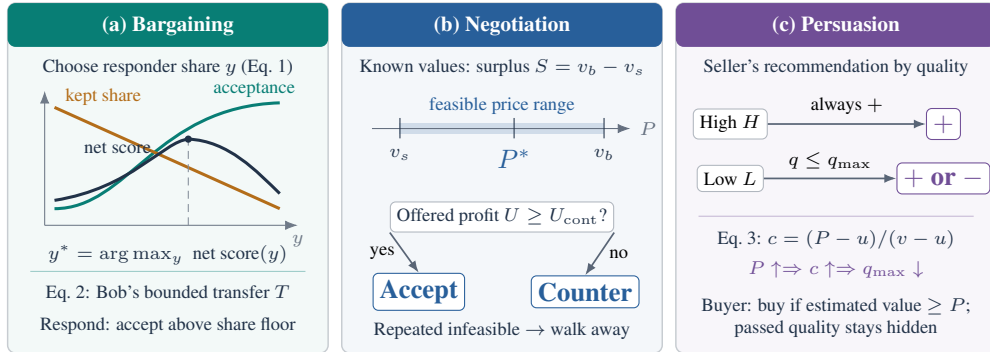


Figure 1: The policy’s three different decision geometries. Curves in (a) are schematic: the bargain search balances modeled acceptance against retained share, then Bob may transfer an additional bounded amount. In (b), price targets lie on the individually rational side of own value and responses compare offered profit with continuation value. In (c), the low-quality + rate is capped only by the static model behind Eq. 3; the executed seller paces signals deterministically, so the figure does not assert history-conditional obedience. All actions pass the legality validator.

Bargaining. For pot M , both roles predict the opponent’s demanded share from a robust median step, blend it with a configuration prior, and search responder shares $y \in \{0.10, 0.105, \dots, 0.94\}$. The score

$$\sigma((y - f + 0.007)/w)(1 - y)^\gamma - [1 - \sigma((y - f + 0.007)/w)]c \quad (1)$$

uses role-specific floor f , width w , curvature γ , and delay cost c listed in Appendix A. The sigmoid estimates acceptance of offering y ; $(1 - y)^\gamma$ values the proposer’s retained share; c penalizes rejection. Alice uses the resulting branch directly. Bob then transfers

$$T = \min\{M[r(\delta_B) + 0.008t], \max(0, x_B - 0.24M)\}, \quad r = 0.018 \text{ if } \delta_B < 0.95 \text{ else } 0.008, \quad (2)$$

where T is a bounded extra transfer from Bob’s share to Alice’s and t is normalized round progress. Bob may also override a rejection when the received share exceeds $\max\{0.16, (0.30 \text{ if } \delta_B < 0.95 \text{ else } 0.33) - 0.09t - 0.055s\}$. These are hand-set heuristics, not Rubinstein equilibrium claims.

Negotiation. Complete-information targets allocate a declining fraction of feasible surplus; hidden-information targets extrapolate half the opponent’s median favorable price step. Offers remain on the individually rational side of own value. Responses compare offered utility with a declining continuation threshold relative to the target price P^* ; repeated profitable offers can be accepted and repeated infeasible unknown-horizon paths can trigger walkaway. Buyer and seller constants differ, and the seller has a short-horizon midpoint repair; Appendix A specifies every branch.

Persuasion. The buyer purchases below low value, passes above high value, and otherwise compares price with a purchased-only pseudo-count reliability score. This is a heuristic, not a Bayesian posterior across heterogeneous games. The seller always recommends high quality. With known buyer values $v > u$, price P , prior p , and a *static randomized* model $\Pr(+ | H) = 1, \Pr(+ | L) = q$, purchase after “+” requires

$$c = (P - u)/(v - u), \quad q \leq q_{\max} = \min\{1, p(1 - c)/[c(1 - p)]\}. \quad (3)$$

ELP sets a target $q = 0.88q_{\max}$ off-terminal and $0.94q_{\max}$ terminal (capped at 0.92), then deterministically signals the n th observed low item when $\lfloor (n + 1)q \rfloor > \lfloor nq \rfloor$. Because q changes and the schedule can be inferred, Eq. 3 is only a calibration heuristic: it does not prove history-conditional obedience. When receiver values are hidden, $q = \text{clip}(0.12 + 0.28p + 0.051_{\text{terminal}}, 0, 0.40)$ is explicitly ad hoc and carries no obedience interpretation.

Validation and exposure. Actions are checked for exact keys, finite values, budget balance, legal enums, required counteroffers, and message length; exceptions invoke a legal fallback. The controller queues one assignment, leaves on assignment, drains the game, and audits authoritative counts and ratings. Family and role loss/drawdown thresholds fixed before the ELP run bound exposure but are not statistical tests.

4 Failure-driven method lineage

Three failures became shared invariants: a legal 99-round cycle motivated repeated-state escape; a 24-game request returning 53 motivated one-assignment recounting; and hidden role losses motivated role-local guards.

Table 1 in Appendix B records branch changes, frozen components, and run coverage. Adaptive defender (AD) adds role-local defense and the guard $R_t < R_0 - 20$ or $R_t < \max_{s \leq t} R_s - 15$. Surplus protection–terminal pooling (SP-TP) uses $f = .75f_0 + .25f_b$ for Alice and gated low-quality signaling. Role-aware reuse–drawdown control (RA-DC) composes $\pi = (\pi_{\text{SP}}^B, \pi_{\text{AD}}^N)$ without a new decision branch.

Adaptive role calibration–contextual concession (ARC) uses $c = c_0 - .04I_{\text{msg}} - .02I_{\text{known, seller}}$. Evidence-guided context repair (EG-CR) blends weak-context negotiation offers toward the midpoint, $p = .85p_0 + .15m$. The submitted evidence-locked portfolio (ELP) uses $q = .88q_{\max}$ and paced low-quality signaling $a_n = \lfloor (n + 1)q \rfloor - \lfloor nq \rfloor$. These are representative branch rules rather than complete method definitions; Appendix A gives the executed ELP specification.

5 Sequential development variants and live trace

The complete dashboard export contains 1,704 games. Summed one-decimal changes were +535.9, +695.0, and +833.0 for bargaining, negotiation, and persuasion. Their per-game mean/median/SD were respectively 1.11/0.0/5.02, 1.24/1.2/4.39, and 1.26/0.9/4.57; sign counts were 240/3/240, 363/0/199, and 361/5/293. Chronological-third gains attenuated from 364.4 to −27.6 to 199.1, 422.8 to 159.0 to 113.2, and 508.1 to 261.7 to 63.2. The platform initializes each family rating at 1,000, so all three final ratings exceed their true platform starts. Displayed gains are non-monotone and attenuate across chronological thirds.

Table 1 adds the historical method baselines. These uncontrolled role-level observations are sequential development variants. Alice’s observed mean was −3.03 under AD ($n = 5$) and +0.43 under

SP-TP ($n = 4$); the retained Bob branch means were $+0.38$ and -1.64 ($n = 6$ each). More strongly, the unchanged defensive negotiation rule averaged $+3.56$ under AD ($n = 50$) but only $+0.24$ under RA-DC ($n = 43$), consistent with substantial temporal and matchmaking variation. Bob’s observed mean was -1.13 under EG-CR ($n = 27$) and $+1.29$ under ELP ($n = 86$); the corresponding persuasion-seller means were -1.58 ($n = 19$) and $+0.51$ ($n = 41$). Negotiation-buyer means remained negative (-1.59 and -1.85), while the frozen seller means were $+1.98$ and -1.49 . The role-isolated rollback, Bob-recovery, and binary-pooling notebooks have executable offline designs but no live outcomes.

ELP then targeted 150 games per family in fixed order at concurrency one. Bargaining reached 150; persuasion stopped at 78 after a 23.19-point drawdown, and negotiation at 20 after a 33.73-point loss. Its family mean/median/SD were 1.27/0.19/4.36, 0.71/ -0.22 /3.90, and -1.69 / -2.98 /3.65. The next-day snapshot ranked the entrant 95th at 1674.3 overall. These are descriptive sequential contrasts: policies were selected from earlier runs, matchmaking and rating drift are uncontrolled, and stopping is outcome-dependent. We report no mechanism-effect estimate or IID interval. Appendix C gives full support, role, dispersion, guard-sensitivity, and execution tables.

6 Agent Behavior Analysis

The executable targets three process properties: individual rationality, escape from repeated paths, and role-separated evidence. Validators test action legality; cycle counters test termination behavior; and family/role guards test exposure. None is a certificate of rating performance. Likewise, the seller’s deterministic frequency target controls aggregate signaling frequency but not dynamic receiver obedience.

The transcript export covers 220 of 248 rated assignments and omits game IDs, so behavior cannot be linked safely to assignment rating. In the 127 available bargaining games, 113 reached agreement (47/54 for Alice and 66/73 for Bob). This measures agreement behavior, not the effect of the recovery override.

In 40 available seller games, ELP positively signaled all 397 high-quality and 135/403 low-quality products. Of 532 positive signals, 397 were high quality (74.6% available-case precision); opposing buyers purchased 309/532 positive- signal and 24/268 negative-signal products. These aggregate responses neither establish counterfactual purchase lift nor verify obedience. In 34 buyer games, ELP purchased 350/680 products (245 high, 105 low); passed quality is hidden, so this purchased subset is selectively revealed. Negotiation reached 11/19 deals, with four no-deals and four walkaways, while both role totals were negative.

The audit’s defensible conclusion is diagnostic: action validity, agreement, signal calibration, and rating success are distinct outcomes. The missing ID linkage prevents attributing rating changes to particular behaviors.

7 Limitations and conclusion

This post-selection, fixed-order trace has uncontrolled matchmaking, role, configuration, opponent, rating shrinkage, and time; two families are adaptively censored. Rating deltas are platform measures, not welfare, and transcripts are incomplete and unlinked. Fixed text cannot test richer dialogue. Strategic persuasion can transfer to manipulative settings despite its calibration-heuristic label and the competition setting.

Bundled V33 evidence regenerates the final-trace tables and figure; the 1,704-game dashboard export is unavailable in this package. Mutable ratings, missing transcripts, and counterfactual actions cannot be replayed. Legal execution, behavior, exposure, and rating effects must be evaluated separately; these totals do not identify mechanism effects.

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A Executable evidence-locked portfolio specification

This appendix transcribes the executed notebook rather than reconstructing the policy from notebook labels. It is pseudocode-equivalent to the six ELP branches; the archived notebook remains authoritative. The definitions below also state the parsing, update, tie-breaking, tolerance, and rounding choices needed to map visible states to actions.

Shared notation and state. $\text{clip}(z, a, b) = \max(a, \min(b, z))$ and $\sigma(z) = (1 + e^{-z})^{-1}$. For known horizon R at round r , $t = (r - 1) / \max(1, R - 1)$; otherwise $t = \min(0.65, (r - 1) / 16)$. A stall count is the minimum length of the two players' trailing approximately repeated offer sequences (relative tolerance 0.003 in bargaining and 0.001 in negotiation). Robust trend is the median adjacent step among the five most recent observations; volatility is their median absolute deviation. Offer and price outputs are rounded to eight decimals. Bargaining threshold comparisons use a 10^{-12} slack and negotiation profitability/threshold comparisons use 10^{-9} .

Memory persists across games only when an opponent name is disclosed. Otherwise the game ID makes it game-local. Bargaining keys additionally include role, information regime, own discount, and visible opponent discount; negotiation keys include role and information regime; persuasion keys include buyer/seller perspective and signal mode. An event set prevents reprocessing history. Thus the action is a deterministic function of visible history, game ID, and this internal memory.

A.1 Bargaining branches

For own player i , let d_j be the opponent shares in their observed offers, $\tilde{d}$ the predicted next demand (last demand plus robust trend), s the stall count, and δ_i own discount. With visible discounts, the configuration prior is $b_0 = 1 - (1 - \delta_j) / (1 - \delta_i \delta_j)$, clipped to $[\cdot001, .999]$; it is .5 when the denominator is at most 10^{-12} [9]. Otherwise it is $0.46 + 0.02t$ against a disclosed human and $0.43 + 0.02t$ otherwise. If z is the largest opponent share in a rejected own proposal (zero before one exists) and m the median of the five latest demands, the base floor used below is $b = \text{clip}(\max\{b_0, z + .006, m - .065 + .020t\}, .20, .999)$, omitting unavailable terms. In the role formulas below, $f_{\text{reject}} = z$.

Alice. If demands exist, set

$$f = \text{clip}\{\max[f_{\text{reject}}, .75b + .25(\tilde{d} - .035 - .020t)], .18, .90\};$$

otherwise $f = \text{clip}(b, .18, .90)$. On an offer turn:

1. If $s \geq 3$, give the responder their last demanded share clipped to $[\cdot20, .82]$ (or $[\cdot20, .94]$ on the final round).
2. Otherwise set $f_s = \text{clip}(f - .010, .18, .88)$, $w = \text{clip}(.016 + .65 \text{MAD}(d), .014, .045)$, and $c = .045 + .25t + .50(1 - \delta_i)$. Search $y = 0.10, 0.105, \dots, 0.94$ and maximize Eq. 1 with $\gamma = 1.18$. Candidate tuples (score, $1 - y, y$) are compared lexicographically, so an exact score tie retains more for the proposer. The responder amount is rounded to eight decimals and the proposer receives the exact residual.
3. Under complete information only, transfer an additional $\min\{M(.018 + .012t), \max(0, x_i - .22M)\}$ from Alice to Bob.

On a response turn, accept a strictly positive final-round offer. Before the final round, accept if $s \geq 3$ and own share is at least .18; otherwise accept when own share is at least

$$\max\{.18, .36 - .08t - .055s, \delta_i(1 - f) \text{clip}(.80 + g + .10t - .06s, .48, .92)\},$$

where the favorable opponent concession is minus the median adjacent step of the recent demand path and $g = \text{clip}(.25 \times \text{favorable opponent concession}, -.02, .02)$. Under complete information, a rejection is finally overridden when own share is at least $\max(.20, .33 - .10t)$.

Bob. Bob uses the same structure with the following replacements: $f = \text{clip}(\max[f_{\text{reject}}, .78b + .22(\tilde{d} - .035 - .020t)], .18, .90)$, $f_s = \text{clip}(f - .006, .18, .88)$, $w = \text{clip}(.016 + .60 \text{MAD}(d), .014, .044)$, $c = .045 + .24t + .50(1 - \delta_i)$, $\gamma = 1.16$, and pre-override risk floor

$\max(.18, .35 - .08t - .055s)$. ELP then applies the offer transfer in Eq. 2. If the base branch rejects, it accepts whenever own share is at least

$$\max\{.16, (.30 \text{ if } \delta_i < .95 \text{ else } .33) - .09t - .055s\}.$$

All proposals divide M exactly. Fixed messages describe the selected split but do not change it.

A.2 Bilateral negotiation branches

Let role $\rho \in \{\text{buyer, seller}\}$, own value v_i , latest opponent price p_o , robust median opponent step Δ , and feasible surplus $S = v_b - v_s$ when both values are visible. Seller utility is $P - v_s$ and buyer utility is $v_b - P$; these are definitions, not an invocation of the Nash bargaining solution.

Buyer target. Under complete information, start with capture $a = .66 - .10t$, add $.35 \text{ clip}(-\Delta/S, -.10, .10)$, subtract $.025 \min(s, 2)(.35 + t)$, and clip a to $[.54, .80]$. Target $P^* = v_b - aS$. Under hidden information with an opponent offer, project the seller's price as $\hat{p} = \max(0, p_o + .5 \min(0, \Delta))$ and set $P^* = v_i - (.64 - .09t)(v_i - \hat{p})$ when $\hat{p} \leq v_i$, otherwise $P^* = v_i$. With no offer, $P^* = v_i(.82 + .09t)$.

Seller target. Under complete information, start with $a = .72 - .12t$, subtract $.04$ when messages are enabled and $.02$ when the horizon is known, add $.35 \text{ clip}(\Delta/S, -.10, .10)$, subtract the same stall term, and clip to $[.54, .80]$. Target $P^* = v_s + aS$. Under hidden information, project a buyer's price as $\hat{p} = \max(0, p_o + .5 \min(0, \Delta))$ and set $P^* = v_i + a(\hat{p} - v_i)$ for $a = .70 - .11t - .041_{\text{messages}}$ when $\hat{p} \geq v_i$, otherwise $P^* = v_i$. With no offer, $P^* = v_i(1.32 - .14t)$.

Responses. Both branches accept a final offer iff its utility is nonnegative. The buyer accepts a profitable repeated path at $s \geq 3$; the seller does so at $s \geq 2$. At $s \geq 2$, a negative-utility path under an unknown horizon causes walkaway. Otherwise compare offered utility U with $|P^* - v_i|k$, where

$$k_{\text{buyer}} = \text{clip}(.76 - .13t - .08 \min(s, 2), .38, .80),$$

$$k_{\text{seller}} = \text{clip}(.80 - .061_{\text{messages}} - .13t - .08 \min(s, 2), .38, .80).$$

Accept if U exceeds that continuation value or, when $S > 0$, captures at least $.52 + .06(1 - t)$ (buyer and no-message seller) or $.48 + .06(1 - t)$ (message seller). Otherwise counter with $(1 - b)P^* + bP$, clipped to the individually rational side, using $b = \text{clip}(.22 + .38t + .08 \min(s, 2), 0, .70)$ for buyer and adding $.10$ with an $.80$ cap for a message-enabled seller.

For complete-information seller contexts with recorded horizon below five rounds (including a missing horizon encoded as zero), ELP additionally blends offers 15% toward the surplus midpoint, accepts nonnegative offers capturing at least $.40 + .08(1 - t)$, and blends rejected counters 18% toward the received price. These historical constants are not estimated equilibrium parameters.

A.3 Persuasion branches

Buyer. Let high/low values be $v > u$, price P , and high-quality prior p . Buy if $P \leq u$ and pass if $P > v$. Otherwise parse the seller signal as positive, negative, or unknown. The parser lowercases text, converts underscores to spaces, removes non-alphanumeric punctuation, and collapses whitespace. Exact negative forms are {no, false, negative, not recommended}; negative phrases are {do not buy, don't buy, should not buy, shouldn't buy, do not recommend, don't recommend, recommend against, not worth, not a good, not high quality, low quality, bad product, buying would be foolish}, and any of {pass, skip, avoid} is negative. These checks precede exact positive forms {yes, true, recommended, buy} and phrases {buy this, recommend buying, recommend this, worth it, high quality, great product, good product, positive}; unmatched text is unknown. For a positive signal use $q_H = .90, q_L = .24$; for a negative signal use $.10, .76$, and first compute $\pi_0 = pq_H/[pq_H + (1 - p)q_L]$. Purchased products update high/low counts; passed products do not update precision because quality is hidden. With $\alpha = 4\pi_0 + n_H$ and $\beta = 4(1 - \pi_0) + n_L$, use the conservative value

$$\hat{p} = \text{clip} \left(\frac{\alpha}{\alpha + \beta} - .55 \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}, 0, 1 \right).$$

For an unknown signal use p . Buy when $\hat{p}v + (1 - \hat{p})u + B \geq P$. Here $\hat{p}$ is only a pseudo-count reliability score: named-opponent memory may pool purchased labels across different priors and prices, so no exchangeable Bayesian model is asserted. The bonus $B = .010(v - u)$ times the remaining-round fraction divided by $\sqrt{1 + n}$ only for a positive signal within $.08(v - u)$ of the cutoff and with fewer than four observed labels; otherwise $B = 0$.

Seller. Always signal positive for high quality. For low quality, compute the target in Eq. 3; return $q = 1$ when $c \leq 0$, $q = 0$ when $c \geq 1$, and $q = 0$ for an interior cutoff with $p \in \{0, 1\}$. When buyer values are hidden use the ad hoc rule reported in the main text. The notebook defines terminal as $r \geq R$ after replacing a missing R with the current round; thus an absent horizon receives the terminal .05 increment and .94 utilization. If n low-quality products precede the current one, signal positive exactly when $\lfloor (n + 1)q \rfloor > \lfloor nq \rfloor$. Text mode emits fixed BUY/PASS sentences and binary mode emits yes/no.

For constant q , the floor-difference sequence has prefix discrepancy below one. ELP recomputes q and reveals signals, so this arithmetic fact supplies no dynamic persuasion or incentive-compatibility guarantee.

A.4 Contract wrapper and exposure controller

The wrapper rejects unexpected keys, nonfinite or negative amounts, allocations that do not sum to the pot within tolerance, illegal decisions, missing nonterminal counteroffers, and messages over 2,000 characters. Its fallbacks are: equal split/accept in bargaining; own-value offer, profitable acceptance, or own-value counter in negotiation; PASS/no as seller; and prior expected-value purchase as buyer. Every fallback is logged.

The controller starts only at zero active games, queues one family, leaves the queue after observing an assignment, drains it, and checks that assignment and authoritative count increments are at most one. Family baseline-loss limits for bargaining/negotiation/persuasion are 30/40/28 and peak draw-downs are 24/30/22 after 20 games. Role limits are 24/28/20 and role drawdowns 18/22/16 after ten role games. Three consecutive matchmaking timeouts stop a family. The executed notebook measured new telemetry before reward harvesting; the 500 caught HTTP 422 errors appended during harvesting therefore escaped the abort check. This historical monitoring-order bug is preserved in the executed artifact and corrected only in the documented future protocol.

Method map. “EG-CR Alice” denotes the Alice branch plus its complete-information correction; “delay-aware Bob” denotes the transfer/acceptance recovery; “AD buyer rollback” denotes the negotiation-buyer equations above; “EG-CR seller” denotes the seller branch plus short-horizon repair; and “AD persuasion buyer” denotes the purchased-only pseudo-count reliability rule. These labels record methodological provenance and are not separate algorithms required to reconstruct ELP.

B Reproducibility and artifact record

Executed agent. The submitted agent is ELP-AGENT, build v33-heuristic-150-each-2026-08-30. The exact executed source is `artifacts/v33/33_evidence_locked_role_portfolio_frequency_paced_bayesian_pooling.ipynb`. It used Python 3.12 and `glee-sdk==0.0.5` in a hosted Kaggle session, concurrency one, no training, no model weights, and no external inference API. The run took approximately 13,497 seconds (3.75 hours); Kaggle’s exact CPU and memory allocation was not recorded.

Bundled evidence. The same directory contains the session report, rating-outcome CSV, policy state, and append-only evidence JSONL. The notebook, policy state, and rating CSV are byte-identical to the executed exports. The evidence replaces only the submitting account UUID and dashboard name with fixed tokens; game states, actions, timestamps, scores, and controller records are unchanged. The rating CSV contains one named synthetic contract-test row and 248 live rows. The JSONL contains ordered decision, move-result, checkpoint (including harvested credit), and family-summary records. File sizes and SHA-256 hashes are listed in `artifacts/`

Table 1: Branch-level sequential development provenance. B/N/P denote bargaining/negotiation/persuasion; outcomes are displayed Δ/n and are uncontrolled historical observations.

Method	Changed branches/controller	Frozen or reused	Live	Observed Δ/n
AD	Alice; negotiation B/S; peak guard	Earlier Bob; persuasion B/S	B,N,P	B $-12.83/11$; N $+177.77/50$; P $-13.85/10$
SP-TP	Alice; P seller; tighter guards	AD Bob; P buyer; N supported, B,P not run	B,N,P	B $-8.13/10$; P $+4.15/10$
RA-DC	Portfolio/guards only; no new decision branch	SP-TP bargaining; exact AD negotiation	B,N	B $+5.86/8$; N $+10.30/43$
ARC	Bob; context-conditioned negotiation B/S	SP-TP Alice; persuasion B/S	B,N,P	B $+10.10/11$; N $+47.41/46$; P $+14.56/16$
EG-CR	Full-information A/B; weak context N B/S; P seller gate	P buyer; remaining contexts	B,N,P	B $+7.81/41$; N $+30.97/68$; P $-11.73/30$
ELP	Bob recovery; N buyer rollback; P seller pacing; wider guards	EG-CR Alice; N seller; P buyer	B,N,P	B $+190.94/150$; N $-33.73/20$; P $+55.67/78$

Table 2: Complete dashboard-history rating statistics. Each family has the platform start of 1,000. Σ sums displayed one-decimal game changes and need not equal Final minus Start because of rounding and platform-wide recalculation.

Family	n	Start	Final	Σ	Mean	Median	SD	+/-
Bargaining	483	1000.00	1535.92	+535.9	1.11	0.0	5.02	240/3/240
Negotiation	562	1000.00	1661.30	+695.0	1.24	1.2	4.39	363/0/199
Persuasion	659	1000.00	1825.84	+833.0	1.26	0.9	4.57	361/5/293

v33/ARTIFACT_MANIFEST.sha256; these hashes, rather than a mutable or identifying repository URL, identify the bundled files. The root-level PACKAGE_MANIFEST.sha256 additionally hashes every bundled source and data file except itself and disposable compiler outputs. The exact AD through EG-CR development notebooks are additionally bundled in artifacts/methods/METHOD_MANIFEST.sha256 records their full hashes and the relative path and hash of the executed ELP notebook.

Development notebook provenance. The notebook filenames and SHA-256 prefixes for the method lineage in Table 1 are: AD, 23_adaptive_defender_peak_drawdown_reservation_protection.ipynb (98904272); SP-TP, 24_bargaining_persuasion_repair_surplus_protection_terminal_pooling.ipynb (9988cecf); RA-DC, 25_role_aware_bargaining_defensive_negotiation_drawdown_control.ipynb (6d203e29); ARC, 26_three_family_adaptive_role_calibration_contextual_concession.ipynb (8995a751); EG-CR, 27_evidence_guided_context_repair_configuration_gated_pooling.ipynb (2dd8d2df); and ELP, 33_evidence_locked_role_portfolio_frequency_paced_bayesian_pooling.ipynb (50248f96). Full hashes and bundled paths are listed in artifacts/methods/METHOD_MANIFEST.sha256.

Regenerating the audit. The final-trace audit can be regenerated with Python 3.12 or later:

```
python -m pip install -r requirements-analysis.txt
python analysis/analyze_v33_transcripts.py
```

This script regenerates the ELP CSVs, tables, JSON summary, and sequential figure from bundled V33 data; it resolves paths from its own location. The complete dashboard-history export and its parser were unavailable in this package, so the 1,704-game history statistics cannot be regenerated from it. The live notebook dependency is separately pinned in requirements-live.txt; live execution additionally requires a valid competition credential and a mutable matchmaking service, and is not needed to verify reported totals.

Replay boundary. Recorded ELP decisions and controller checkpoints can be audited from the append-only evidence. We do not claim complete counterfactual replay: 28 human-readable transcripts are absent, that export lacks IDs, the opponent population and rating model evolve, and

Table 3: Leaderboard snapshot supplied by the author. The player label is the study’s agent alias.

Rank	Player	Type	Rating	Bargaining	Negotiation	Persuasion
95	ELP-agent	Agent	1674.3	1535.9	1661.3	1825.8

unobserved alternative actions do not exist. Exact leaderboard ratings and missing transcript behavior therefore cannot be reproduced. The policy action map can be inspected from the executable and the states that are present in the evidence.

Known executed defect. The 500 telemetry rows are caught HTTP 422 failures for six stale component-test IDs. Synthetic classification was overwritten or incomplete, and the controller measured new telemetry before harvesting. The defect did not alter recorded actions or ratings, but it prevented the intended telemetry abort. It is not silently patched in the executed notebook; Appendix C specifies the corrected future ordering.

C Supplementary statistical and systems audit

Sources, order, and linkage. The author-reported complete dashboard history comprises 1,704 rated entries from August 24–30: 483 bargaining, 562 negotiation, and 659 persuasion. The underlying export and parser are unavailable in this package, so these descriptive summaries cannot be independently regenerated from it. Four games were against humans and remain in the reported history. The platform starts each family rating at 1,000. Summed displayed changes are rounded to one decimal and need not equal the difference between this start and the higher-precision final rating; platform-wide recalculation can also prevent exact additivity.

Chronological thirds sum to $+364.4/ - 27.6/ + 199.1$ for bargaining, $+422.8/ + 159.0/ + 113.2$ for negotiation, and $+508.1/ + 261.7/ + 63.2$ for persuasion. Long-run role means were $+1.11$ Alice, $+1.11$ Bob, $+0.67$ negotiation buyer, $+1.84$ negotiation seller, $+1.71$ persuasion buyer, and $+0.86$ persuasion seller. These partitions describe attenuation and role asymmetry across changing policies and opponents; they are not stationarity tests or estimates of version effects. The family and role statistics for this complete history are not machine-readable in the supplied package.

The rating analysis uses all 248 live rows in `data/v33_rating_outcomes.csv` after excluding the explicitly named synthetic contract-test row. File order is the executed assignment order; the analysis writes `data/v33_rating_outcomes_sequenced.csv` with an explicit 1–248 index. The append-only evidence and session report independently retain checkpoint order and assigned game IDs.

Table 3 records the separate author-supplied August 31 dashboard snapshot; it is not reconstructed from assignment deltas.

The text export contains 220 complete records: 127/150 bargaining, 74/78 persuasion, and 19/20 negotiation (88.7% overall). Two records omit a repeated header but retain separate setup and result blocks, which the parser recovers. The export has no game IDs, so transcript records are not positionally joined to ratings. No missing-at-random assumption is made; behavioral quantities are available-case process summaries. For example, the 113 observed bargaining agreements imply only the worst-case full-run range 113/150–136/150 (75.3–90.7%) without assumptions about the 23 missing records.

Descriptive rating paths. Table 5 reports signs, means, medians, dispersion, and no confidence interval. The positive bargaining mean coexists with an almost even sign split; persuasion has a positive mean but negative median. Lag-one sample autocorrelations were 0.05, -0.08 , and 0.07 for bargaining, persuasion, and negotiation, but do not establish independence. Repeated opponents cannot be identified from the rating extract. Consequently neither a fixed-sample IID interval nor a post hoc confidence sequence would have a clear estimand or justified dependence structure.

Figure 2 plots cumulative changes against the executed family guards. Bargaining’s maximum within-run drawdown was 24.75 points before the 20-game activation point. Persuasion crossed its 22-point boundary at game 78; negotiation crossed 30 points at game 20.

Table 4: Executed safeguard and repair activations. “Units” counts distinct games, or families for the drawdown row.

Mechanism	Units	Activations	Detail
Contract fallback	0	0	of 1,838 decisions
Bob recovery override	38	46	29 offers; 17 accepts
Cycle intervention	3	3	1 match; 2 walkaways
Seller short-horizon repair	3	6	4 from unknown horizon
Family drawdown stop	2	2	persuasion; negotiation
Assignment overshoot	0	0	across 248 checkpoints
Telemetry abort	0	0	500 errors after checks

Table 5: Descriptive distribution of assignment-level displayed rating changes. No inferential interval is reported because matchmaking, opponent repetition, drift, and stopping are uncontrolled.

Family	+ / 0 / −	Mean	Median	SD	IQR
Bargaining	76/1/73	1.27	0.19	4.36	7.64
Persuasion	38/0/40	0.71	-0.22	3.90	6.62
Negotiation	7/0/13	-1.69	-2.98	3.65	6.16

Support and confounding. Table 6 gives the full role–mode–value–visibility support. Every hidden-value persuasion assignment placed ELP in the seller role. Hidden-versus-known marginal totals therefore combine information and role effects and are not compared. The machine-readable context CSV retains role, information, horizon, discount, signal mode, prior, and cutoff buckets; small cells remain exploratory and no multiplicity-adjusted claims are made.

Available-case behavior. Among 127 exported bargaining games, 113 reached agreement and 14 ended in walkaway; agreements used a median one observed round (mean 1.40, maximum 12). Negotiation produced 11/19 agreements, four no-deals, and four walkaways. The 74 persuasion games contain 1,480 decisions. In 40 seller games, 397/397 observed high-quality products and 135/403 low-quality products received a positive signal. Opponents bought 309/532 positive-signal and 24/268 negative-signal products. In 34 buyer games, ELP bought 350/680 products, including 245 high and 105 low; passed quality is hidden. None of these unlinked summaries estimates a rating effect or counterfactual behavior.

Guard sensitivity. Table 9 replays the observed sequence while jointly scaling all family and role thresholds. At half thresholds, bargaining would have stopped at game 17, persuasion at 24, and negotiation at 19. At 1.25 times the thresholds, neither adaptively stopped family crosses a guard in its observed prefix. These are path-dependent replays, not counterfactual estimates: later matchmaking need not remain the same after a different stopping decision.

Executed horizons and safeguards. All 78 persuasion games recorded 20 rounds, so its missing-horizon terminal branch was never activated. Horizons were unknown in 76 bargaining games (113 decisions) and 10 negotiation games (28 decisions). One complete-information seller game with an unknown horizon triggered the short-horizon repair on all four of its decisions; two further repair activations occurred in known short-horizon games. A future policy should use a separate unknown-horizon branch rather than encode it as zero.

The evidence-level audit in `data/v33_execution_activation_audit.csv` counts 1,838 live policy decisions. Table 4 distinguishes safeguards that fired from those merely present in code.

Telemetry defect. The session report contains 500 caught HTTP 422 reward-harvest failures from six non-UUID component-test IDs. Two IDs were queried 84 times and four were queried 83 times. The ELP wrapper overwrote three synthetic flags, three other IDs did not match the synthetic prefix classifier, and the checkpoint measured new telemetry before harvesting. Hence the abort guard saw zero newly appended errors. The queries occurred after actions and did not change recorded ratings, but they invalidate any claim that the monitoring layer worked as intended. The future protocol must preserve synthetic flags, classify test IDs by an explicit registry, and measure telemetry after harvesting.

Table 6: Persuasion role–mode–information support. Hidden-value assignments contain no ELP buyers, so hidden/known marginal differences are not role-adjusted comparisons.

Role	Mode	Values	n	Sum	Mean	+ / 0 / −
Buyer	binary	known	16	11.63	0.73	8/0/8
Buyer	text	known	21	23.14	1.10	14/0/7
Seller	binary	hidden	11	-2.07	-0.19	3/0/8
Seller	binary	known	9	4.79	0.53	4/0/5
Seller	text	hidden	15	-3.93	-0.26	5/0/10
Seller	text	known	6	22.11	3.68	4/0/2

Table 7: Role-stratified assignment-level rating changes.

Family	Role	n	Sum	Median	SD	+ / 0 / −
Bargaining	Alice	64	80.15	0.25	3.82	33/1/30
Bargaining	Bob	86	110.79	-0.08	4.74	43/0/43
Persuasion	Buyer	37	34.77	0.73	3.89	22/0/15
Persuasion	Seller	41	20.90	-0.45	3.95	16/0/25
Negotiation	Buyer	11	-20.33	-2.91	4.02	4/0/7
Negotiation	Seller	9	-13.40	-3.04	3.37	3/0/6

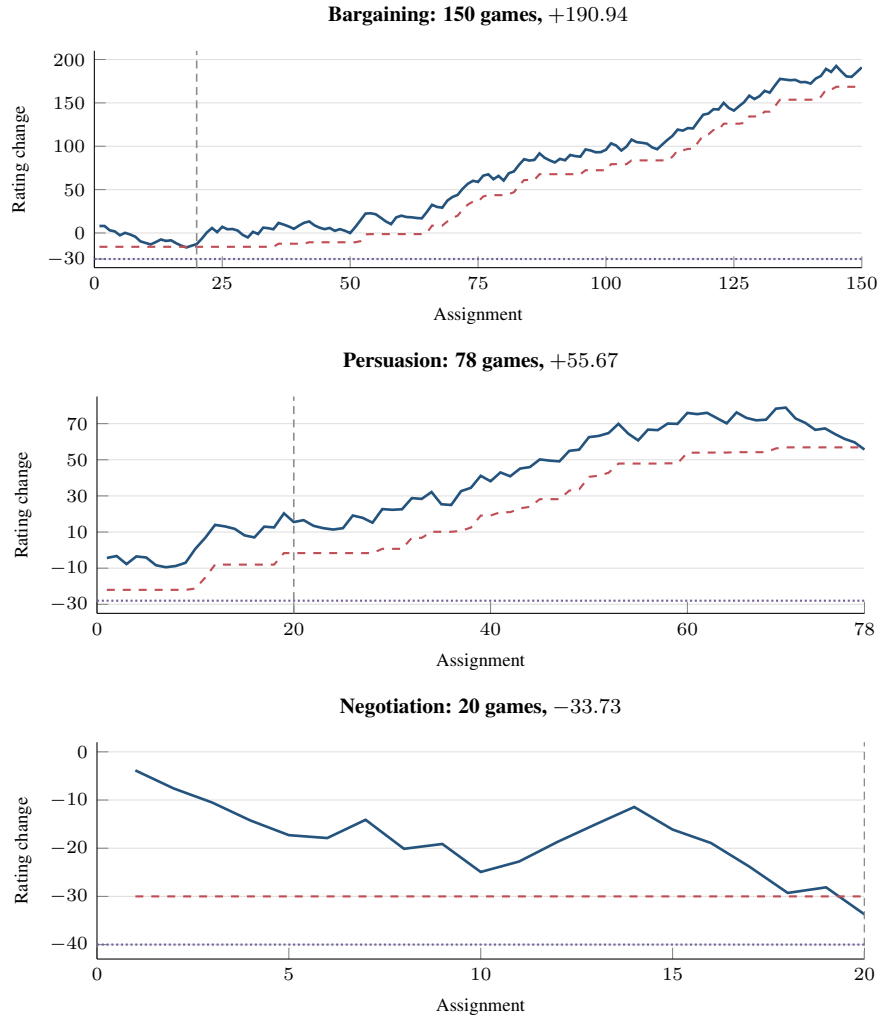


Figure 2: Cumulative displayed rating changes from the ordered rating CSV. Solid blue is cumulative change; dashed coral is running peak minus the family drawdown limit; dotted purple is the baseline-loss limit. The vertical gray line marks the 20-game guard activation point. These controller boundaries are not inferential thresholds.

Table 8: Outcome counts in the transcript export. Coverage is 220 of 248 rated assignments, so these are available-case behavioral summaries and are not joined to rating rows.

Family	Role	n	Agreement	Walkaway	No deal
Bargaining	alice	54	47	7	0
Bargaining	bob	73	66	7	0
Persuasion	buyer	34	0	0	0
Persuasion	seller	40	0	0	0
Negotiation	buyer	11	6	3	2
Negotiation	seller	8	5	1	2

Table 9: Stop-guard sensitivity when all four loss/drawdown thresholds are multiplied by a common factor. Entries are simulated stop game and reason from the observed family sequence.

Family	0.50×	0.75×	1.00×	1.25×	1.50×
Bargaining	17 (Bob loss)	17 (Bob DD)	150 (target)	150 (target)	150 (target)
Persuasion	24 (buyer DD)	75 (buyer DD)	78 (family DD)	78 (end)	78 (end)
Negotiation	19 (buyer loss)	20 (buyer DD)	20 (family DD)	20 (end)	20 (end)

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